F957 — Two holds, both worked. The load-bearing one (Cal's separation catch) resolves clean: the reproducing exponent 5/2 and the up-quark Yukawa mode-weight 7 are distinct objects, so the up-quark data does *not* kill 5/2. And the curved-sea stitch is set up, with an honest note on where the real curvature constant will decide things. ★ The separation check (Cal's catch) — resolved: distinct, by variance. Two different quantities were being conflated: (A) the reproducing-kernel exponent s — the analytic weight of the fermionic projector P(x,y) = S(B/G)·G^{−s}, the short-distance singularity of the two-point function, set by the spinor bundle $\Lambda^{\{0,*\}\otimes K}${1/2}; this is universal — the *same* for the electron, up, charm, top, because it is the bundle, not the fermion; value genus/2 = n_C/rank = 5/2. (B) the up-quark Yukawa mode-weight N — y_up = n_C^{−N}, the *coupling* of one specific K-type mode to the Higgs; this is flavor-specific — y_e ≠ y_up ≠ y_top, so N varies by flavor (up: N=7). A universal number and a flavor-varying number cannot be the same quantity. So 5/2 (reproducing) and 7 (up-Yukawa) coexist without conflict — the up-quark data does not kill 5/2. Cal's catch was the right question, and the answer is two objects, not one. (Had they been the same, the data would kill 5/2 at the flavor scale — load-bearing, not bookkeeping. They're not.) ★ 5/2 is the *curved* weight, not a flat artifact. Keeper flagged "the 5/2 was on the flat one." Clarifying: the 5/2 was derived on F947's Bergman-operator kernel, which *is* curved — B(x,y) depends on the base point (F956). And det S(B/G) = 1 (F953), so the spinor factor carries no G-power on *or* off the geodesic slice ⍰ s is purely the K^{1/2} bundle weight = genus/2 = 5/2, on the curved kernel. So 5/2 is the leading reproducing exponent on the curved operator (curvature corrections are subleading and don't move the leading exponent). The flat build (F954) is a *different* object (correct Clifford/Γ_5/Krein, no curvature) and was never the weight-carrying kernel. ★ The mode-weight 7 — a distinct object, its own derivation, blind (setup, not closed by me). y_up = n_C^{−7} is the up-quark's flavor-specific coupling — a K-type localization overlap (Paper 118 Sec 9.3, per-flavor assignment). The "7" is the up-quark's mode-step count; blind candidates: 7 = g (signature), or 7 = N_c + rank^2 = 3+4. This is Elie's K1201 / the fermion-assignment domain, multi-month; I confirm only that it is a *different* object from the reproducing 5/2 and must be derived on its own terms. ★ The curved-sea stitch — set up; the pieces are verified, uniting them needs the curved operator. The synthesis: F947 (curved reproducing kernel, curvature via B, weight 5/2) + F954 (Γ_5 chirality = the Krein grading, on the real operator) + F952 (Lorentzian causal structure, spacelike L=0 / timelike real, verified) + mass. Each piece is verified separately; the honest hold is that uniting them into one curved-Lorentzian-graded massive fermionic projector requires the covariant curved Dolbeault-Dirac (∂̄† with the varying Bergman

The separation, in one line

	reproducing exponent (A)	up-Yukawa mode-weight (B)
what	two-point-function singularity (bundle)	Higgs coupling of one K-type (fermion)
depends on flavor?	NO (universal, 5/2)	YES (up: 7)
verdict	distinct objects — 5/2 survives; data does not touch it	derive on its own terms (Sec 9.3)

The two honest holds (Keeper's framing)

1. The sea is still flat — the built operator (F954) is flat; the curved kernel (F947) exists; the stitch = the covariant curved operator (Sec 9.1). Elie's 8.50-vs-8.75 measures whether curvature is in.
2. 5/2 is provisional-on-the-curved-operator — confirmed 5/2 is the *leading* curved reproducing weight (K^{1/2}, det S=1); the full massive/Dirac-numerator kernel is checked by the m→0 arbiter (which needs the curvature in first — itself the diagnostic that it isn't yet).

RUBRICS Layer-2 done-bar (two holds, honest)

- ☒ Resolved Cal's separation catch: 5/2 (universal reproducing) ≠ 7 (flavor-specific Yukawa), by variance; the up-quark data does not kill 5/2. Load-bearing, clean.
- ☒ Confirmed 5/2 is the *curved* reproducing weight (F947's B is curved; det S=1 → no extra G-power → K^{1/2} bundle weight), not a flat artifact.
- ☒ Set up the mode-weight 7 as a distinct object (flavor-specific, Sec 9.3, Elie's domain); did not close it, confirmed only the separation.
- ☒ Set up the curved-sea stitch (F947+F954+F952+mass into the covariant curved Dolbeault-Dirac, Sec 9.1); flagged it as the multi-week integral, not a re-derivation.
- ☒ Mapped Elie's c-discriminator to $|\rho|^2$=8.50 (no g) vs Lichnerowicz 8.75 (g=7); committed to neither. Nothing pushed.

Handoffs

- @Cal — your separation catch resolves to distinct. 5/2 = universal bundle exponent (two-point singularity, flavor-independent); 7 = flavor-specific up-quark Yukawa (K-type coupling). A universal and a flavor-varying number can't be one object → 5/2 survives, data doesn't kill it. Hostile question back: do you accept "universal vs flavor-specific" as a genuine separation, or is there a channel where the reproducing exponent *does* feed the flavor Yukawa (mixing the two)? And: 5/2 confirmed curved (det S=1 on the curved B), so the reproducing check is "5/2 on the curved kernel," 35/4 dropped.
- @Elie — your c-discriminator is the sharpest diagnostic, and here's the map. The zero-momentum curvature constant is one number: 8.50 = $|\rho|^2$ = (n_C^2+N_c^2)/4 (no g) vs 8.75 = n_C·g/4 (Lichnerowicz, g=7). We dropped 35/4 (=8.75) because it imports g=7; so 8.50 is the g-free-consistent value — but measure it blind, you've committed to neither. The m→0→P_exact_positive(s=2.5) check can't run until the curvature's in — which, as you said, is itself the diagnostic that it isn't yet. Build the covariant $\bar\partial^\dagger$ (varying Bergman metric, Sec 9.1) → the curvature constant falls out → then the sea + census.
- @Keeper — two holds worked. (1) Separation resolved (Cal's catch): 5/2 (universal reproducing) and 7 (flavor Yukawa) are distinct by variance → 5/2 survives, up-quark data doesn't kill it. (2) 5/2 confirmed as the *curved* weight (F947's B is curved; det S=1 → K^{1/2} bundle), not a flat artifact — the "flat" concern was that F954 (flat) isn't the weight kernel; F947 (curved) is. The stitch = the covariant curved Dolbeault-Dirac (Sec 9.1), into which F947+F954+F952+mass plug — the multi-week integral, honestly the hold. Elie's 8.50-vs-8.75 is the go/no-go on curvature.
- @Grace — 5/2 survives (distinct from the up-Yukawa 7). Register the DE forward-falsifier at teeth: complete-monotonicity forbids a crossing in the *geometric* H(z), so a crossing in the *direct* Hubble rate would falsify BST by our own kill condition — that's a real forward falsifier, not a fit. SNe-side residual

analysis earns it when the data's in hand. #76 structural parts closed; the Peirce statement is Lyra/Cal-deep (I'll take it with Cal after the sea).

- @Casey — the number I was worried about survived, for an honest reason. Cal asked the sharp question: is the fermion's "weight" (five-halves) secretly the same thing as the up-quark's coupling power (seven)? Because if they were the same, the up-quark's measured mass would *kill* five-halves outright. I checked, and they're genuinely different animals — one is a property of the *space* (the same for every particle), the other is a property of *one particular particle* (different for the electron, the up-quark, the top). A number that's the same for everyone and a number that changes particle-to-particle can't be the same number. So five-halves lives; the seven in the up-quark is its own separate question, for its own derivation. And there's a clean single-number test coming: when Elie builds the curved version, one constant will read either 8.50 or 8.75 — and the 8.75 is the one that smuggles a seven back in, so we're watching for 8.50, the honest g-free value. He measures it blind. The sea is still flat and I've said so plainly; the curved object exists and the last mile is one genuine integral. Nothing pushed.

Notes only; no toy/theorem claimed (separation resolution + stitch setup). F957: SEPARATION CHECK (Cal) RESOLVED — DISTINCT by variance. (A) reproducing exponent s = analytic weight of P(x,y)=S(B/G)G^{-s} = spinor BUNDLE $\Lambda^{\{0,*\}\otimes K}${1/2} weight = UNIVERSAL (same all fermions) = genus/2 = n_C/rank = 5/2. (B) up-quark Yukawa mode-weight N: y_up=n_C^{-N}, K-type COUPLING = FLAVOR-SPECIFIC (up N=7). Universal≠flavor-varying → distinct → up-quark data does NOT kill 5/2. 5/2 CONFIRMED CURVED: F947's B(x,y) curved (F956), det S=1 (F953) → no G-power → s=K^{1/2} bundle weight=5/2 on curved kernel, not flat artifact (F954 flat = different object, not weight kernel). MODE-WEIGHT 7 = distinct object, own derivation blind (7=g? or N_c+rank^2=7?), Paper 118 Sec 9.3 fermion-assignment = Elie K1201 domain, NOT closed by me. STITCH: curved sea = F947 (curved kernel wt 5/2) + F954 (Γ_5 Krein grading) + F952 (Lorentzian causal) + mass, uniting = covariant curved Dolbeault-Dirac (∂† varying Bergman metric, Sec 9.1) = multi-week integral, the hold. ELIE c-DISCRIMINATOR mapped: zero-momentum curvature constant = 8.50=$|\rho|^2$=(n_C^2+N_c^2)/4 (NO g) vs 8.75=n_C·g/4 Lichnerowicz (g=7); dropped 35/4=8.75 imports g=7 → 8.50 g-free-consistent, but Elie measures blind. Cal: accept universal-vs-flavor separation? 5/2 curved. Elie: build covariant ∂† → curvature constant 8.50/8.75 blind → m→0→P_exact_positive(s=2.5). Keeper: 2 holds worked, 5/2 survives+curved, stitch=Sec9.1 integral. Grace: 5/2 survives; DE falsifier teeth (monotone H(z) forbids crossing). Nothing pushed. — Lyra